

ProSinger: Zero-shot Technique Controlled Natural Singing Voice Synthesis

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Abstract

Zero-shot singing voice synthesis (SVS) with style transfer and style control aims to generate high-quality singing voices with unseen timbres and styles (including singing methods, emotions, and techniques) from audio and text prompts. However, existing style-controllable models often suffer from poor naturalness due to data scarcity or architectural deficiency, while large-scale style-transfer models achieve superior fidelity but lack mechanisms for fine-grained, phoneme-level manipulation. To address these limitations, we propose **ProSinger**, a framework capable of generating both natural singing voices and providing precise phoneme-level technique control. Specifically, ProSinger utilizes an autoregressive transformer to generate content-style tokens, followed by a flow-matching transformer to produce acoustic representations. Our innovation is primarily composed of two parts: 1) we introduce **Technique Tokens**—one-hot representations that encode six distinct singing techniques (falsetto, glissando, mixed voice, pharyngeal, breathy, and vibrato) for every individual phoneme—enabling the autoregressive stage to generate content-style tokens guided by explicit technique instructions; 2) by fine-tuning an existing large-scale pre-trained autoregressive transformer, ProSinger inherits a robust generative prior and superior naturalness while gaining localized expressive flexibility. Experimental results demonstrate that ProSinger effectively achieves high-fidelity singing voice synthesis with phoneme-level technique controllability.

Index Terms: singing voice synthesis, singing style control

1. Introduction

Singing Voice Synthesis (SVS) aims to generate high-quality singing voices from provided lyrics and musical notations. Recent advancements in the field have shifted focus toward zero-shot SVS, which encompasses two primary sub-tasks: style control and style transfer. Style control aims to generate singing with fine-grained controllability for style, while style transfer focuses on generating singing with global style prompts. Modern frameworks in these tasks often decompose singing into three primary attributes: linguistic content (what to sing), timbre (who sings), and style (how to sing). Achieving targeted and controllable imitation of these attributes is essential for applications ranging from personalized music production to cross-lingual dubbing.

A major challenge in current SVS research is the trade-off between synthesis naturalness and fine-grained controllability. On one hand, self-supervised style-transfer frameworks, such as Vevo[1] and Vevo2[2], have demonstrated outstanding natu-

ralness in zero-shot scenarios. By leveraging a decoder-only autoregressive (AR) transformer for content-style modeling and a flow-matching transformer for acoustic modeling, these models achieve high-quality singing synthesis through large-scale pre-training. However, their control remains largely at the global level, lacking the mechanism for precise, phoneme-level manipulation of singing techniques.

On the other hand, specialized multi-level style control models like TCSinger[3] have successfully modeled diverse singing expressions by predicting style embeddings via a dedicated language model. This allows users to specify phoneme-level techniques. Despite this flexibility, these systems often suffer from poor naturalness due to architectural complexity or the scarcity of high-quality annotated training data.

To bridge this gap, we propose ProSinger, a novel framework that introduces fine-grained controllability into the high-fidelity architecture Vevo2. Rather than introducing a new acoustic modeling stage, we focus on specializing the **autoregressive content-style transformer**. Recognizing the superior generative priors of large-scale pre-training, we perform fine-tuning on a robust pre-trained AR checkpoint to incorporate phoneme-level expressiveness. Our core innovation is the introduction of **Technique Tokens** into the AR transformer stage. These tokens encode specific singing techniques (e.g., mixed voice, falsetto, breathy, vibrato, glissando, pharyngeal) for every individual phoneme, enabling the generation of content-style tokens that are guided by explicit technique instructions rather than solely by implicit audio references.

The contributions of this work are summarized as follows:

- We introduce Technique Tokens to encode six distinct singing behaviors for every individual phoneme: falsetto, glissando, mixed voice, pharyngeal, breathy, and vibrato.
- By fine-tuning the pre-trained autoregressive transformer, our framework inherits the robust naturalness of large-scale pre-training while gaining the stylistic versatility required for professional singing synthesis.
- We propose a pipeline to train technique-controlled singing voice synthesis model unsupervisedly, which is easier to gain better performance with dataset scaled up.
- Experimental results demonstrate that ProSinger matches the synthesis quality of state-of-the-art singing voice synthesis systems while significantly outperforming existing models in local style controllability and expressiveness.

2. Proposed Algorithm: ProSinger

In this section, we demonstrate how we specialize Vevo2 for precise technique controlled singing voice synthesis by introducing our proposed technique tokenizer, inference pipeline,

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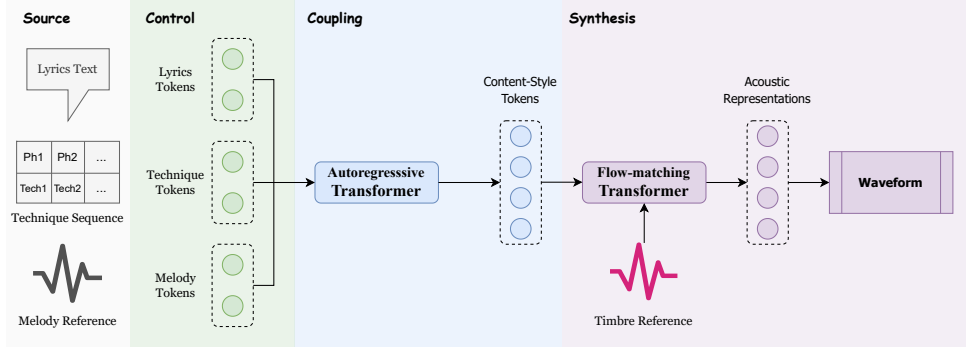


Figure 1: The inference pipeline of ProSinger.

and fine-tuning strategy.

2.1. Technique Tokenizer

The original Vevo2 model utilizes an additional singing audio clip as the global style reference. While it can automatically infer singing techniques from the reference audio, precise control of local singing technique can rarely be achieved. To address this, we proposed a straightforward technique tokenizer. It takes phoneme-aligned technique sequences as input and encodes the technique combination for each phoneme into a one-hot token. Given k technique types, this representation create a token space of size 2^k , allowing for fine-grained, phoneme-level control.

2.2. ProSinger

As illustrated in Figure 1, ProSinger’s core design follows three fundamental questions: what to sing (lyric texts), how to sing (technique sequence), and who is singing (singer timbre). We split the whole pipeline into three stages:

Control Stage The system employs three fixed tokenizers: a pre-defined text tokenizer, the proposed technique tokenizer, and a pre-trained melody tokenizer, to encode corresponding input sources into discrete tokens.

Coupling Stage The concatenated tokens serve as the conditioning input to an autoregressive transformer. This module predicts a sequence of content-style tokens, which form a robust hybrid representation that jointly encodes the linguistic content and the singing style dictated by the technique sequence.

Synthesis Stage Conditioned on the content-style tokens and the acoustic features of the timbre reference, a Flow-matching Transformer generates the corresponding acoustic representations (e.g., mel-spectrograms). These are subsequently decoded into the final singing waveform.

2.3. fine-tuning Autoregressive Transformer

To adapt the Vevo2 architecture for explicit technique control, we make a key modification to its autoregressive transformer. Specifically, we substitute the original reference content-style tokens in the input with our phoneme-aligned technique tokens. Consequently, the model is trained to directly predict the target content-style tokens conditioned on these explicit technique instructions, rather than inferring them implicitly from a reference audio.

We initialize the transformer with the pre-trained weights from Vevo2. For the newly introduced embeddings in the tech-

nique token vocabulary, we initialize them with the mean vector of the pre-existing token embeddings. This strategy promotes stable training onset by leveraging the model’s prior knowledge of relevant feature distribution.

Demonstrated in Figure 2, the fine-tuning process is fully self-supervised. For each training singing audio clip, we first extract and decouple it into three parallel condition sources: lyrics, technique sequence (using either human annotations or pseudo-labels from a pre-trained detector), and melody. These conditions are then encoded by their respective tokenizers. Concurrently, the same original audio clip is processed by the pre-trained content-style tokenizer to generate the target content-style tokens. Following common practice in discrete sequence modeling, we employ the cross-entropy loss for this prediction task.

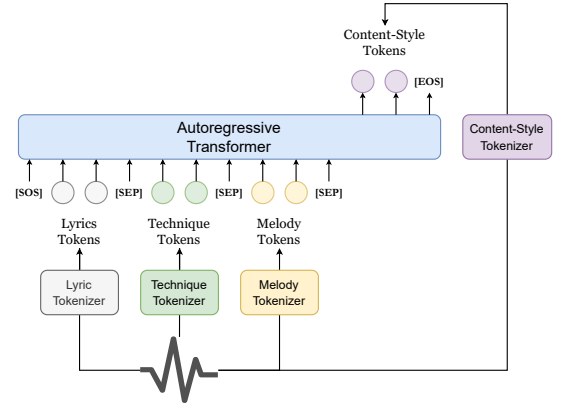


Figure 2: Finetune Autoregressive Transformer. Only Autoregressive Transformer’s parameters are unfrozen during the fine-tuning process.

3. Experiments

We train and evaluate ProSinger on an extended dataset, containing Chinese subset of GTSinger[4], which has human-annotated phoneme-level technique labels, and extended data based on other singing voice datasets using pseudo-label tagging. All experiments are conducted on a single RTX 5880 Ada GPU.

Table 1: *Objective comparison on FPC, WER, and TAR.*

Model	FPC $\uparrow$	WER $\downarrow$	TAR $\uparrow$
TCSinger	0.796	1.790	0.386
Vevo0.4k	0.776	0.605	–
ProSinger(finetune)	0.833	0.485	0.422

3.1. Datasets

Our method requires singing technique sequences for supervised learning. However, singing technique annotation requires professional threshold and a large amount of manpower. Thus, except for GTSinger, we utilize STARS[5], a publicly accessible tool for automatic singing voice annotation, to obtain pseudo-labels for phoneme-level technique. Using this method, we extended the total Chinese singing dataset from 50 hours to more than 200 hours.

GTSinger, a dataset containing 80.9 hours of singing voice, with controlled comparison and phoneme-level annotations of six commonly used singing techniques. We use the technique groups as our training data, while the control (no technique) groups are used in the evaluation.

3.2. Experiment Results

We proposed three objective metrics to evaluate model’s performance, including F0 correlation(FPC), Word Error Rate(WER), and Technique Accuracy Rate (TAR), to assess the consistency of melody, lyrics, and techniques between generated audio and ground truth. For Technique Accuracy Rate, we use ProSinger and GTSinger to generate audio, and use STARS to recognize the phone-level techniques of synthesized singing, then calculate the accuracy rate with ground truth labels. For FPC and WER, we compare the performance of ProSinger, Vevo1.5 0.4k and GTSinger. Result shows that ProSinger perform much better than TCSinger in style control, while maintaining high capability to follow melody and lyrics.

3.3. Result Demonstration

We build a demo page to show the effectiveness of ProSinger along with its comparison with two baseline systems: TCSinger and Vevo2 at <https://techprosinger.github.io/>.

4. Conclusions

In this paper, we propose ProSinger, a novel zero-shot singing voice synthesis framework that achieves fine-grained, phoneme-level control over singing techniques while maintaining high naturalness. The core innovation is the introduction of explicit technique tokens into the autoregressive content-style coupling stage, enabling users to precisely dictate localized singing behaviors. To ensure audio fidelity, our approach inherits the robust generative prior and generalization capability from large-scale pre-trained models. Taken together, ProSinger effectively bridges the gap between global style transfer and local expressive control in singing synthesis.

5. Future Plan

While ProSinger achieves promising results, we identify several key areas for further enhancement:

- **Monotonic Attention for Technique Alignment:** Current autoregressive modeling relies on standard attention mecha-

nisms which may occasionally lead to alignment shifts. We plan to implement **monotonic attention** to strictly enforce the temporal correspondence between phonemes and their associated technique tokens. This will ensure that expressive behaviors are accurately localized without skipping or repeating, mirroring the stable temporal dependencies observed in AR speech models.

- **Mitigating Pitch Ambiguity by Replacing Chromagram Features:** To further improve the robustness of the synthesis stage, we aim to add one more dimension to chromagram features to inform model with octave. We have identified that chromagram features tend to unify all pitches within a single octave, effectively discarding the absolute register information. This folding mechanism often causes the model to confuse the actual pitch during generation, leading to octave errors. By adding one more dimension to represent octave, model will be able to receive and output precise pitch information during the process.

6. References

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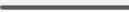
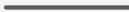
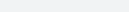
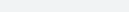
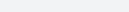
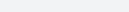
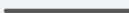
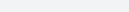
Methods	Lyrics	Technique Prompts	Results
Vevo	时间都去哪了还没好好感受年轻就老了	▶ 0:00 / 0:09  🔊 ⋮	▶ 0:00 / 0:09  🔊 ⋮
TCSinger		Breathy for the first half, then Pharyngeal for the second half, with Vibrato at the end.	▶ 0:00 / 0:09  🔊 ⋮
ProSinger (Ours)			▶ 0:00 / 0:09  🔊 ⋮
Methods	Lyrics	Technique Prompts	Results
Vevo	怕你飞远去怕你离我而去	▶ 0:00 / 0:08  🔊 ⋮	▶ 0:00 / 0:08  🔊 ⋮
TCSinger		Pharyngeal for the first half, then Breathy and Falsetto for the second half.	▶ 0:00 / 0:07  🔊 ⋮
ProSinger (Ours)			▶ 0:00 / 0:07  🔊 ⋮
Methods	Lyrics	Technique Prompts	Results
Vevo	菊花残满地伤你的笑容已泛黄	▶ 0:00 / 0:09  🔊 ⋮	▶ 0:00 / 0:09  🔊 ⋮
TCSinger		Breathy for the first half, then Falsetto for the second half, with Vibrato for the whole sentence.	▶ 0:00 / 0:08  🔊 ⋮
ProSinger (Ours)			▶ 0:00 / 0:09  🔊 ⋮

Figure 3: Demo page at <https://techprosinger.github.io/>.